SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS:50 **Total Marks:25**

Annexure A

Multiple Choice Questions (MCQ)

Code of Conduct:

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student**: M. Pranay*

Annexure A

Short Answer Test

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?
   a) To compute only prior probabilities
   b) To update prior beliefs using new evidence
   c) To eliminate irrelevant features
   d) To maximize classification accuracy

2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which of the following statements correctly describes this assumption in practical machine learning applications?
   a) Features are completely dependent on each other
   b) Features are conditionally independent given the class
   c) Features must be continuous
   d) Features are always correlated

3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which of the following options correctly explains the objective of MLE when fitting a model to observed data?
   a) Maximize prior probability
   b) Minimize hypothesis space
   c) Maximize likelihood of observed data
   d) Reduce computational complexity

4. The Minimum Description Length principle is often used in model selection. Which of the following best explains how MDL balances model complexity and goodness of fit in learning algorithms?
   a) Prefers highly complex models always
   b) Prefers simplest model that explains data well
   c) Ignores data noise completely
   d) Focuses only on training accuracy

5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which of the following statements best describes how it achieves the minimum possible classification error?
   a) By selecting random hypotheses
   b) By averaging predictions over all hypotheses
   c) By using only prior probabilities
   d) By minimizing dataset size

6. The Gibbs Algorithm is an important concept in Bayesian learning. Which of the following options best explains how it selects a hypothesis from the hypothesis space during classification?

a) Selects hypothesis randomly from consistent set
b) Selects hypothesis with minimum loss only
c) Selects fixed predefined hypothesis
d) Ignores hypothesis space

7. Naïve Bayes classifiers are often used in text classification tasks. Which of the following conditions ensures that the classifier performs effectively despite its simplifying assumptions?
a) Strong dependence between features
b) Large amount of labeled data
c) Features being conditionally independent
d) Absence of noise in dataset

8. Bayesian Belief Networks are graphical models that represent probabilistic relationships. Which of the following best describes the role of nodes and edges in such networks?
a) Nodes represent variables and edges represent dependencies
b) Nodes represent datasets and edges represent features
c) Nodes represent outputs only
d) Edges represent noise in data

9. The Expectation-Maximization algorithm is commonly used when dealing with incomplete data. Which of the following best explains the role of the Expectation step in this iterative algorithm?
a) Updates model parameters
b) Computes expected values of hidden variables
c) Removes missing data
d) Sorts the dataset

10. In the EM algorithm, the Maximization step updates certain parameters. Which of the following correctly describes what is updated during this step to improve model likelihood?
a) Hidden variables only
b) Observed data values
c) Model parameters maximizing likelihood
d) Training dataset size

11. Probability learning involves estimating probability distributions from data. Which of the following best explains the goal of probability learning in the context of machine learning models?
a) To remove all uncertainty
b) To estimate underlying probability distributions
c) To reduce number of features
d) To simplify model structure

12. Sample complexity is an important concept in learning theory. Which of the following best describes what sample complexity indicates about a learning algorithm's performance?
a) Amount of computation required
b) Number of samples needed for learning

c) Model accuracy only
d) Memory consumption

13. A finite hypothesis space contains a limited number of possible hypotheses. Which of the following best explains how this limitation impacts the learning process and generalization ability?
a) Improves flexibility always
b) Restricts possible models considered
c) Eliminates need for data
d) Guarantees perfect accuracy

14. In contrast to finite hypothesis spaces, infinite hypothesis spaces allow more flexibility. Which of the following best describes the characteristics of such hypothesis spaces in learning models?
a) Limited to discrete functions
b) Allows continuous and complex models
c) Contains no hypotheses
d) Simplifies computation always

15. The mistake bound model is used to evaluate learning algorithms. Which of the following best describes what is measured by the mistake bound in online learning scenarios?
a) Training time
b) Maximum number of mistakes made
c) Memory usage
d) Dataset size

16. Prior probability is a key concept in Bayesian inference. Which of the following best describes what prior probability represents before observing any new evidence?
a) Probability after evidence
b) Initial belief before observing data
c) Conditional probability only
d) Joint probability

17. Posterior probability is calculated after incorporating evidence. Which of the following best describes how posterior probability is derived using Bayes' Theorem?
a) Using only likelihood
b) Using prior and likelihood with evidence
c) Ignoring prior information
d) Using random sampling

18. Likelihood plays a central role in statistical inference. Which of the following best explains what likelihood represents in relation to observed data and hypotheses?
a) Probability of hypothesis given data
b) Probability of data given hypothesis
c) Probability of prior only
d) Probability of error

19. Naïve Bayes classifiers are widely used in real-world applications. Which of the following best describes the type of problems where this algorithm is most effective?

a) Clustering problems
b) Classification problems with high-dimensional data
c) Regression problems
d) Sorting problems

20. Overfitting is a common issue in machine learning models. Which of the following best describes a situation where a model is considered to be overfitting the training data?
a) Model performs well on unseen data
b) Model captures noise instead of pattern
c) Model is too simple
d) Model ignores data

21. Underfitting occurs when a model fails to capture patterns in data. Which of the following best describes the characteristics of an underfitted model in practice?
a) High accuracy always
b) Fails to learn patterns from data
c) Too complex model
d) Overtrained model

22. The EM algorithm is iterative and requires a stopping condition. Which of the following best explains when the algorithm is considered to have converged?
a) When data changes
b) When parameters stabilize across iterations
c) When iterations increase
d) When accuracy decreases

23. Bayesian learning combines prior knowledge with observed data. Which of the following best describes how these two components interact during the learning process?
a) Prior replaces data
b) Data replaces prior
c) Both are combined to update belief
d) Neither is used

24. Concept learning involves identifying a suitable hypothesis. Which of the following best explains the main goal of concept learning in machine learning systems?
a) To reduce computation
b) To find hypothesis consistent with data
c) To eliminate features
d) To increase dataset size

25. Hypothesis space search is a key component of concept learning. Which of the following best describes how searching through hypothesis spaces helps in selecting an appropriate model?
a) Random selection of hypothesis
b) Exploring possible hypotheses to find best fit
c) Ignoring data patterns
d) Reducing dataset size

26. In Bayesian inference, normalization ensures probabilities sum to one. What role does the evidence term play in Bayes' Theorem?
a) It acts as a scaling factor
b) It removes prior probability
c) It increases model complexity
d) It selects hypothesis randomly

27. In Naïve Bayes classification, what is the primary reason for using logarithms when computing probabilities?
a) To increase accuracy
b) To avoid numerical underflow
c) To remove dependencies
d) To simplify dataset

28. Which of the following best describes conditional probability?
a) Probability of two events occurring together
b) Probability of an event given another event has occurred
c) Probability of independent events
d) Probability of random variables

29. Which factor most affects the performance of a Naïve Bayes classifier in real-world datasets?
a) Number of classes
b) Independence assumption violation
c) Dataset size only
d) Output format

30. In Bayesian networks, what ensures that the model remains computationally efficient?
a) Large dataset
b) Conditional independence
c) Continuous variables
d) High dimensionality

31. Which of the following best describes the role of prior knowledge in Bayesian learning?
a) Eliminates need for data
b) Guides learning before observing data
c) Reduces accuracy
d) Replaces hypothesis

32. In Maximum Likelihood Estimation, what happens when the dataset size increases significantly?
a) Prior dominates
b) Likelihood estimation becomes more accurate
c) Model becomes random
d) Complexity decreases

33. Which of the following is a limitation of MLE?
a) Uses too much prior knowledge
b) Ignores prior information

c) Cannot handle data
d) Always overfits

34. Which of the following best describes Bayesian estimation compared to MLE?
a) Uses only likelihood
b) Uses both prior and likelihood
c) Ignores uncertainty
d) Works only for small datasets

35. What is the main advantage of Laplace smoothing in Naïve Bayes?
a) Removes features
b) Handles zero probability problem
c) Reduces dataset size
d) Increases bias only

36. Which of the following best defines a hypothesis in machine learning?
a) Final output
b) Assumed model for mapping inputs to outputs
c) Dataset
d) Feature vector

37. What is the purpose of hypothesis space in learning algorithms?
a) Store data
b) Define all possible models
c) Reduce training time
d) Increase features

38. Which of the following best explains generalization in machine learning?
a) Memorizing training data
b) Performing well on unseen data
c) Reducing dataset size
d) Increasing parameters

39. What is the effect of noisy data on learning algorithms?
a) Improves accuracy
b) Leads to overfitting
c) Eliminates features
d) Reduces complexity

40. In EM algorithm, what is the main challenge during initialization?
a) Dataset size
b) Choosing initial parameters
c) Output format
d) Removing features

41. Which of the following best describes convergence in iterative algorithms?
a) Increasing error
b) Stabilization of parameters
c) Random updates
d) Dataset reduction

42. Which of the following is a characteristic of probabilistic models?
a) Deterministic outputs
b) Handling uncertainty
c) Ignoring data
d) Fixed predictions

43. What is the role of likelihood in Bayesian inference?
a) Represents prior
b) Represents evidence
c) Measures fit of data to hypothesis
d) Removes uncertainty

44. Which of the following best describes overfitting?
a) Model too simple
b) Model fits noise in training data
c) Model ignores data
d) Model has low variance

45. Which of the following best describes underfitting?
a) Model too complex
b) Model fails to capture patterns
c) Model memorizes data
d) Model has high variance

46. What is the role of training data in learning algorithms?
a) Defines output only
b) Guides model learning
c) Reduces complexity
d) Eliminates hypothesis

47. Which factor improves model generalization?
a) Increasing noise
b) Proper regularization
c) Reducing data
d) Ignoring validation

48. Which of the following best explains feature selection?
a) Removing dataset
b) Selecting relevant features
c) Increasing features
d) Ignoring attributes

49. Which of the following best describes classification problems?
a) Predict continuous values
b) Assign labels to data
c) Cluster data
d) Reduce dimensions

50. Which of the following best describes regression problems?
a) Predict discrete labels
b) Predict continuous values

c) Cluster data
d) Classify data

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Answers

1. Bayes' Theorem
Correct answer: B
To update prior beliefs using new evidence.
Explanation: Bayes' Theorem is used to update the probability of an event when new information is available. It combines prior probability with new evidence to calculate posterior probability.

2. Naïve Bayes Classifier
Correct answer: B
Features are conditionally independent given the class.
Explanation: Naïve Bayes assumes that features are independent of each other when the class is known. This assumption makes classification simple and fast.

3. Maximum Likelihood Estimation (MLE)
Correct answer: C
Maximize likelihood of observed data.
Explanation: MLE finds the best model parameters by selecting values that make the observed data most likely to occur.

4. Minimum Description Length (MDL)
Correct answer: B
Prefers the simplest model that explains data well.
Explanation: MDL selects a model that balances simplicity and accuracy. It avoids unnecessarily complex models and helps prevent overfitting.

5. Bayes Optimal Classifier
Correct answer: B
By averaging predictions over all hypotheses.
Explanation: The Bayes Optimal Classifier considers all possible hypotheses and combines their predictions based on their probabilities. This helps achieve minimum expected classification error.

6. Gibbs Algorithm
Correct answer: A
Selects hypothesis randomly from consistent set.
Explanation: The Gibbs algorithm selects a hypothesis randomly according to its probability distribution and uses it to make predictions.

7. Naïve Bayes Performance
Correct answer: C
Features being conditionally independent.
Explanation: Naïve Bayes works effectively when features are conditionally independent given the class. This assumption simplifies probability calculations.

8. Bayesian Belief Networks
Correct answer: A
Nodes represent variables and edges represent dependencies.
Explanation: In a Bayesian network, nodes represent variables and directed edges show relationships or dependencies between variables.

9. Expectation Step (E-Step)
Correct answer: B
Computes expected values of hidden variables.
Explanation: The E-step estimates the expected values or probabilities of hidden variables using the current model parameters.

10. Maximization Step (M-Step)
Correct answer: C
Model parameters maximizing likelihood.
Explanation: The M-step updates model parameters to maximize the expected likelihood calculated during the E-step.

11. Probability Learning
Correct answer: B
To estimate underlying probability distributions.
Explanation: Probability learning estimates probability distributions from available data to understand uncertainty and make predictions.

12. Sample Complexity
Correct answer: B
Number of samples needed for learning.
Explanation: Sample complexity refers to the number of training examples required by an algorithm to learn a model with the desired accuracy.

13. Finite Hypothesis Space
Correct answer: B
Restricts possible models considered.
Explanation: A finite hypothesis space contains a limited number of possible models. The learning algorithm searches among these models to find a suitable hypothesis.

14. Infinite Hypothesis Space
Correct answer: B
Allows continuous and complex models.
Explanation: An infinite hypothesis space contains unlimited possible hypotheses. It provides more flexibility to represent different patterns and relationships.

15. Mistake Bound Model
Correct answer: B
Maximum number of mistakes made.
Explanation: The mistake bound model measures the maximum number of incorrect predictions made by a learning algorithm during online learning.

16. Prior Probability
Correct answer: B
Initial belief before observing data.
Explanation: Prior probability represents the initial belief about an event before considering new evidence.

17. Posterior Probability
Correct answer: B
Using prior and likelihood with evidence.
Explanation: Posterior probability is calculated using Bayes' Theorem. It combines prior probability and likelihood after observing new evidence.

18. Likelihood
Correct answer: B
Probability of data given hypothesis.
Explanation: Likelihood measures how probable the observed data is under a particular hypothesis or model.

19. Naïve Bayes Applications
Correct answer: B
Classification problems with high-dimensional data.
Explanation: Naïve Bayes is commonly used for classification tasks involving many features, such as spam detection, text classification and sentiment analysis.

20. Overfitting
Correct answer: B
Model captures noise instead of pattern.
Explanation: Overfitting occurs when a model learns the training data too closely, including unnecessary noise. As a result, it performs poorly on unseen data.

21. Underfitting
Correct answer: B
Fails to learn patterns from data.
Explanation: Underfitting occurs when a model is too simple to understand the important patterns in the training data. It performs poorly on both training and testing data.

22. EM Algorithm Convergence
Correct answer: B
When parameters stabilize across iterations.
Explanation: The EM algorithm is considered converged when the model parameters or likelihood values stop changing significantly between iterations.

23. Bayesian Learning
Correct answer: C
Both are combined to update belief.
Explanation: Bayesian learning combines prior knowledge with observed data to calculate posterior probabilities and improve predictions.

24. Concept Learning
Correct answer: B
To find a hypothesis consistent with data.
Explanation: Concept learning aims to identify a hypothesis or rule that correctly describes the target concept using training examples.

25. Hypothesis Space Search
Correct answer: B
Exploring possible hypotheses to find best fit.
Explanation: Hypothesis space search examines different possible models and selects a suitable hypothesis that best matches the training data.